WEEK 10 PROJECT REPORT

REAL-ESTATE ANALYTICS DASHBOARD

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Project Title: Real-Estate Analytics Dashboard using Power BI
Tool Used: Microsoft Power BI Desktop
Dataset: King County House Sales Dataset

INTRODUCTION

The Week 10 Power BI project focuses on Real-Estate Visualization and Dashboard Development. The primary objective of this task was to transform the structured real estate semantic data model generated during Week 9 into a fully interactive, production-ready enterprise executive dashboard.

To accomplish this, custom Data Analysis Expressions (DAX) metrics were developed within a dedicated schema folder, and advanced coordinate-based maps, category distributions, and conditional feature impact components were systematically layered onto a high-contrast dark theme canvas. Furthermore, responsive dual-handle numeric parameters were engineered to empower end-users with real-time on-canvas data filtering.

PREVIOUS STAGE WORKFLOW (Week 9 Summary)

During Week 9, the raw King County housing dataset was ingested and sanitized inside the Power Query Editor. Anomalous database rows containing unfeasible zero-bedroom or zero-bathroom parameters were programmatically removed. The text-based date timestamps were standardized into standard calendar constraints, enabling the extraction of Sale Year and Sale Month. Finally, custom dimension variables were engineered specifically Property Age at Sale, Price per SqFt, and Total Area, alongside ordered logical grouping columns (Size Bucket and Price Tier) and multi-clause binary renovation markers (Is Renovated). This curated data layout established the clean foundation required to drive the calculations and visual objects built this week.

CORE TASK DELIVERABLES & TECHNICAL IMPLEMENTATIONS

1. Formulation of Dedicated DAX Measures

- **Total Properties Sold:** Total Properties Sold = COUNTROWS(kc_house_data)
- **Average House Price:** Avg House Price = AVERAGE(kc_house_data[price])
- **Median Price:** Median Price = MEDIAN(kc_house_data[price])
- **Average Living Area:** Avg Living Area = AVERAGE(kc_house_data[sqft_living])
- **Average Price per Square Foot:** Avg Price/SqFt = AVERAGE(kc_house_data[Price per SqFt])

2. High-Level KPI Summary Cards

Five individual **Card (123)** visuals were lined up systematically down the left margin of the page to serve as primary visual anchor blocks. These components tap directly into the custom DAX measure schemas to provide stakeholders with an instant, real-time macro overview of the active dataset matrix.

- **Total Properties Sold Card:** Displays a rounded whole value of **22K** transaction records.
- **Avg House Price Card:** Formatted as Currency ($) showing an active mean position of **$540.20K**.
- **Avg Living Area Card:** Formatted as a Whole Number indicating an interior spatial baseline of **2.08K** square feet.
- **Avg Price/SqFt Card:** Formatted as Currency showing a rate metric baseline of **$264.11**.



3. Spatial and Geographic Real Estate Mapping

An interactive **Map** object was embedded into the bottom-left layout zone of the canvas to convert coordinate pairs into visible market distribution markers. After adjusting the desktop security

preference parameters under *Options > Global > Security* to authorize real-time internet data connectivity, the visual was mapped to Bing's spatial geocoding infrastructure.

- **Latitude Field:** Controlled by the spatial lat dataset property.
- **Longitude Field:** Controlled by the spatial long dataset property.
- **Bubble Size / Formatting:** Linked directly to the calculated Avg House Price measure, rendering dynamic coordinate groupings that visually distinguish high-value luxury coastal clusters from dense, lower-cost interior neighborhoods.



4. Cross-Sectional Pricing Distribution Visualizations

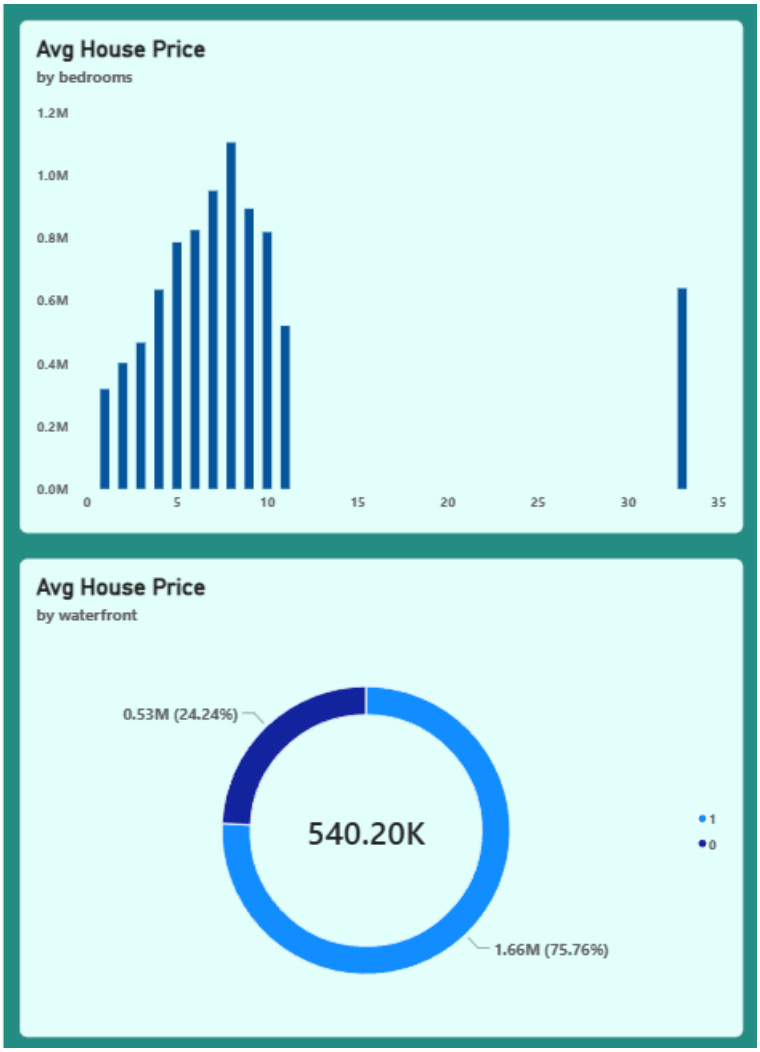
To evaluate how specific structural variables, environmental premiums, and qualitative upkeep features influence overall asset values, a series of multi-axis charts were integrated across the right-hand quadrants:

Price Distribution by Bedroom Count

- *X-Axis:* Continuous bedrooms capacity numbers.
- *Y-Axis:* Calculated Avg House Price metric, exposing the correlation between room count extensions and scaling house prices.

Premium Waterfront Value Impact

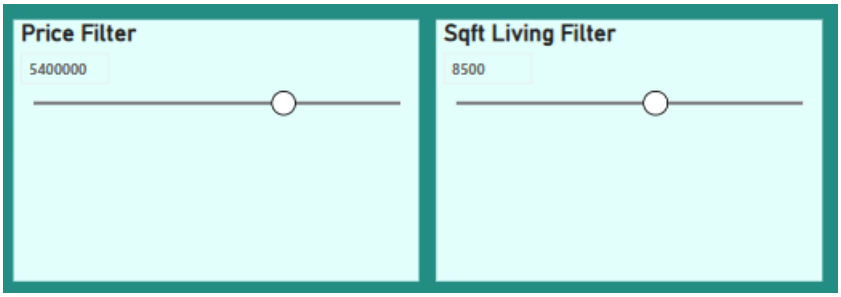
- *Legend Grouping:* Split by binary waterfront indicators (0 = Standard Property, 1 = Waterfront Asset).
- *Values Field:* Mapped to Avg House Price to isolate the massive real estate valuation premiums commanded by waterfront real estate.



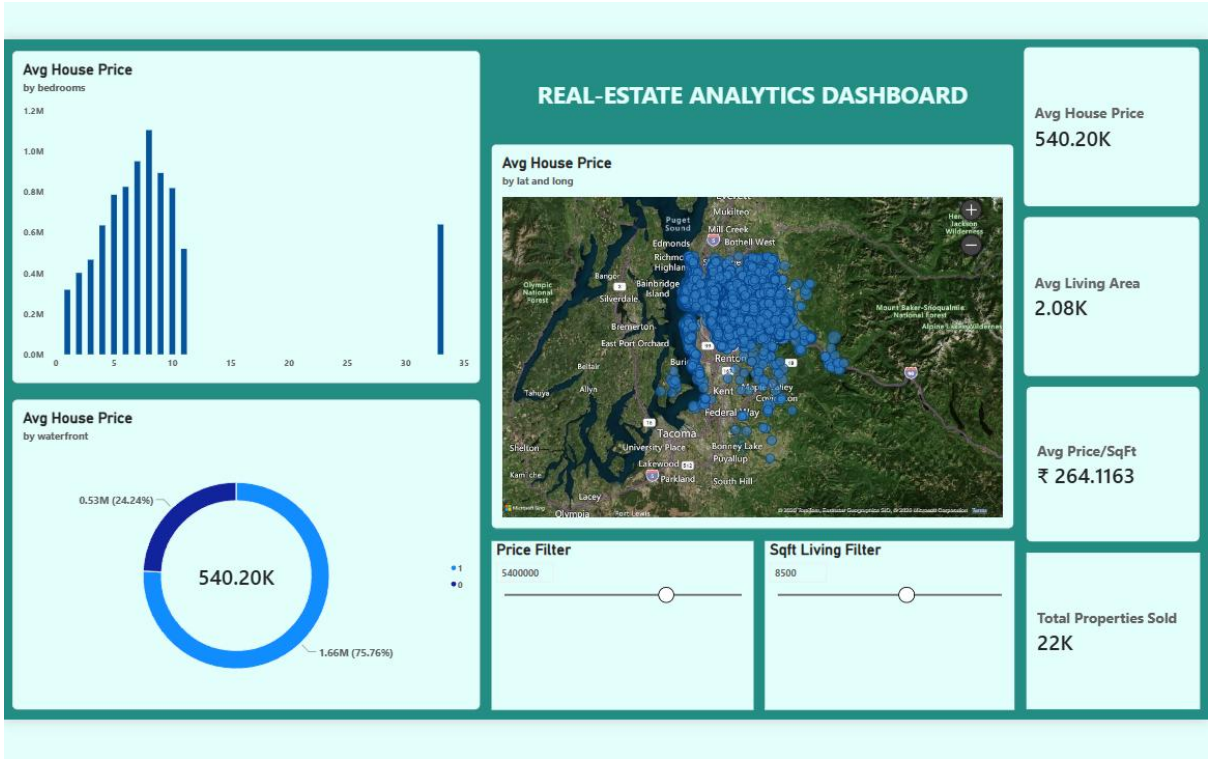
5. Dynamic Range Parameters & Slicer Modeling

To give end-users total exploratory control over the real estate models, custom user-control parameters were introduced via the top **Modeling** ribbon pane using the **Numeric Range** engine tool.

- **Price Filter Slider Container:** A custom numerical slider range named Price Filter. Configured with a Minimum of 0, a luxury scale Maximum of 8,000,000, and a slider step Increment of 50,000.
- **Square Footage Filter Slider Container:** A custom range filter named Sqft Living Filter. Built using a Minimum size constraint of 0, a spatial Maximum limit of 15,000, and an interaction step Increment of 250.



FINAL DASHBOARD LAYOUT



CONCLUSION

The Week 10 project provided extensive hands-on experience in mathematical DAX programming, geographic spatial integration, premium parameter tracking, and interactive cross-filtering dashboard architecture inside Power BI Desktop.